AI Experiments Document

Smart Campus Assistant

CyberPro AI-Powered Development Bootcamp | ELAD Software

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1. Introduction

This document presents the results of AI experiments conducted on the Smart Campus Assistant. The experiments evaluate the system's accuracy, response quality, and prompt .engineering effectiveness across different question categories and system prompt versions

The experiments cover: 10 test questions across all categories, comparison between two system prompt versions (v1 and v2), identification and analysis of three failure cases, and .performance metrics including response time and confidence calibration

2. Experiment Setup

2.1 System Configuration

LLM Model: gpt-4o-mini (OpenAI) | Embedding Model: text-embedding-3-small (1536 dimensions)

Vector Database: Supabase with pgvector | RAG Top-K: 5 chunks | Confidence Threshold: 0.7

Dataset: 16 campus data records across 3 categories (schedule, general_info, technical_issue)

2.2 System Prompt Versions

Version 1 (v1) - Basic Instructions

Provides basic role assignment and rules. Instructs the model to answer only from provided context, be concise, and respond in the same language. Uses a simple 4-tier confidence scale .from 0.0 to 1.0

Version 2 (v2) - Professional with Safety Focus

Adds explicit safety rules (ignore role-change attempts), uses labeled sections (CONTEXT, STRUCTURED DATA, FORMAT), and provides a refined confidence calibration guide that .distinguishes between 'directly stated' and 'clearly supported' answers

2.3 Test Questions

#	Question	Expected Category	Type
1	What are the office hours ?for Dr. Cohen	schedule	Standard
2	?When is the Python exam	schedule	Standard
3	?Where is the cafeteria	general_info	Standard
4	I can't log into the student portal	technical_issue	Standard
5	How do I connect to ?campus WiFi	technical_issue	Edge case
6	What time does the ?library close	general_info	Standard
7	Tell me a joke	out_of_scope	Out-of-scope
8	Hebrew: office hours for Dr. Levi	schedule	Bilingual
9	What is the meaning of ?life	out_of_scope	Out-of-scope
10	How do I set up VPN for ?remote access	technical_issue	Standard

3. Test Results - Prompt Version 2 (Active)

#	Question	Category	.Conf	Time	Result	Sources
1	Office hours Dr. Cohen	schedule	85%	9605ms	Correct	campus_db
2	Python exam	schedule	85%	4998ms	Correct	campus_db
3	Cafeteria location	general_info	85%	5123ms	Correct	campus_db
4	Portal login	technical_issue	85%	6261ms	Correct	Portal Login
5	Campus WiFi	technical_issue	30%	-	Fallback	-
6	Library closing	general_info	85%	4122ms	Correct	campus_db
7	Tell me a joke	out_of_scope	-	-	Blocked	-
8	Hebrew (Dr. Levi)	-	-	-	Error 422	-
9	Meaning of life	out_of_scope	-	-	Blocked	-
10	VPN setup	technical_issue	85%	5062ms	Correct	VPN Setup

V2 Summary: 7/10 successful, 1 correct fallback (WiFi 30%), 2 out-of-scope blocks, 1 error .(Hebrew). Avg response time: 5,737ms. Avg confidence: 0.85

4. Test Results - Prompt Version 1

#	Question	Category	.Conf	Time	Result	Sources
1	Office hours Dr. Cohen	schedule	100%	10090ms	Correct	source1
2	Python exam	schedule	100%	5045ms	Correct	source1
3	Cafeteria location	general_info	100%	4613ms	Correct	campus_db
4	Portal login	technical_issue	100%	5647ms	Correct	campus_db
5	Campus WiFi	-	-	-	Not tested	-
6	Library closing	general_info	100%	5691ms	Correct	campus_db
7	Tell me a joke	out_of_scope	-	-	Blocked	-
8	Hebrew (Dr. Levi)	-	-	-	Error 422	-
9	Meaning of life	out_of_scope	-	-	Blocked	-
10	VPN setup	technical_issue	100%	5871ms	Correct	VPN Setup

V1 Summary: 6/10 successful, 2 out-of-scope blocks, 1 error (Hebrew), 1 not tested. Avg response .time: 6,160ms. All answers returned 100% confidence (overconfidence issue)

5. Prompt Version Comparison

5.1 Quantitative Comparison

Metric	V1 (Basic)	V2 (Professional)
Average Confidence	(100%) 1.00	(85%) 0.85
Confidence Calibration	Poor - always 100%	Good - reflects partial context
Avg Response Time	6,160ms	5,737ms
Source Attribution	Generic ('source1')	Specific (titles, campus_db)
Safety Rules	Basic rules only	Explicit safety section
Fallback Triggered	Never (overconfident)	Yes - WiFi at 30%

5.2 Analysis

Confidence Calibration: V1 returns 100% confidence on every answer regardless of context quality. This defeats the fallback mechanism. V2 returns 85%, reflecting that answers are inferred from context rather than directly stated. This enables the fallback to work as .designed

Source Attribution: V1 returns generic labels ('source1') with no traceability. V2 returns meaningful names ('Student Portal Login Issues', 'VPN Setup for Remote Access'), allowing .students to verify information

Answer Quality: Both versions produce accurate, concise answers. The factual content is identical because both use the same RAG context. The difference is in metadata quality .(confidence, sources) rather than answer text

5.3 Conclusion

V2 is the recommended production prompt. Its superior confidence calibration means the fallback mechanism works correctly, and its source attribution improves transparency. V1 is unsuitable for production because its 100% confidence prevents the system from ever .admitting uncertainty

6. Failure Cases - Analysis

6.1 Failure #1 - Hebrew Question (JSON Encoding Error)

Question: Hebrew text asking about Dr. Levi's office hours

Expected: Answer about Dr. Levi's office hours (schedule category)

Actual: HTTP 422 - JSON decode error

Root Cause: The Hebrew quotation marks inside the question contain a double-quote character that breaks JSON parsing in Swagger UI. The question never reaches the classifier - it .is a client-side encoding issue

Fix: The frontend chat UI handles this correctly via JavaScript's native JSON.stringify(). The .issue only occurs in Swagger. For robustness, adding input encoding middleware would help

Lesson: Always test through the actual frontend, not just Swagger. The frontend's .escapeHtml() and JSON.stringify() handle special characters correctly

6.2 Failure #2 - WiFi Question (Low Confidence Fallback)

?Question: How do I connect to campus WiFi

Expected: WiFi connection instructions (technical_issue)

Actual: Fallback response at 30% confidence

Root Cause: The classifier correctly categorized as 'technical_issue', but the WiFi data is stored under 'general_info' category in the seed data. The metadata filter excluded it. The two-pass .fallback found WiFi info but with low semantic similarity

Fix: Re-categorize WiFi seed data as 'technical_issue', or add dual-category tagging. The two-pass retrieval strategy (SRS 9.5) correctly caught this but the semantic gap between question .phrasing and stored content was too large

Lesson: Category boundaries between general_info and technical_issue are ambiguous. Cross-.category topics need dual tagging or broader search

6.3 Failure #3 - V1 Overconfidence (100% on All Answers)

Question: All questions tested with prompt v1

Expected: Varying confidence scores (0.7-1.0)

Actual: Every answer returned confidence 1.0 (100%)

Root Cause: V1's confidence guide says '0.9-1.0: Direct answer found in context'. The LLM interprets any relevant context as justification for maximum confidence. V1 lacks nuance to .'distinguish 'directly stated' from 'reasonably inferred

Fix: Resolved in V2 with granular confidence guide: '0.85+: directly stated' vs '0.70-0.85: .clearly supported'. The lower ceiling and behavioral descriptions produce calibrated scores

Lesson: Confidence calibration is a prompt engineering challenge. Telling the LLM 'rate 0-1' .without precise anchoring leads to overconfidence

7. Performance Metrics

Metric	V1	V2
Fastest response	4,613ms (cafeteria)	4,122ms (library)
Slowest response	10,090ms (Dr. Cohen)	9,605ms (Dr. Cohen)
Average	6,160ms	5,737ms
?Under 3s target (NFR-01)	No	No

Both versions exceed the 3-second target (NFR-01). The bottleneck is 3 sequential OpenAI API calls (classify, embed, generate). Future optimizations: cache frequent embeddings, use a local .classifier, and implement async parallel execution for RAG and structured data queries

8. Conclusions & Recommendations

Prompt v2 is recommended for production due to superior confidence calibration and source .1 .attribution

The fallback mechanism works as designed - WiFi at 30% correctly triggered staff contact .2 .instead of a wrong answer

.Out-of-scope detection is robust - both versions correctly blocked non-campus questions .3

.Hebrew support requires frontend-side encoding, not AI changes .4

.Category boundaries need refinement - cross-category topics like WiFi need dual tagging .5

Response time optimization is the top future priority - reducing sequential API calls .6
.through caching or local models